CHAPTER 6
CONCLUSION AND FUTURE WORK

6 Conclusion

InterviewSign is a mobile application that helps Deaf, Hard of Hearing and Speech Impaired job seekers to practise interviews in a way that suits them. Most interview preparation tools expect spoken answers. InterviewSign allows the user to answer by typing text or by recording a sign language video, and then gives instant AI feedback and a score. In this way the user can prepare for an interview without depending on speech.

The application is built with React Native and Expo using TypeScript, and it uses Supabase for authentication, the PostgreSQL database and file storage. The user can register and log in with an email or a username, choose an interview round, a technology or focus area and the topics, answer the questions, and review every attempt in the History screen. The Home screen shows the attempts, the average score and the streak, and the Profile screen keeps the personal details and the profile photograph. Protected navigation, Row Level Security and the 7-day username rule keep the accounts and the data safe.

All the objectives of the project were achieved. The testing described in Chapter 4 covered 34 test cases, 20 unit tests and 14 integration tests, and all of them passed. The results in Chapter 5 show that the screens are simple, the cards and icons are large and clear, and the scores are shown in colour, so the user can understand the progress without reading long text. A single code base runs on both Android and iOS, which makes the application available to many users.

During the project, practical skills were gained in mobile application development, file-based navigation, state management with React Context, database design, security rules and user interface design for accessibility. InterviewSign shows that a simple and accessible tool can help candidates with hearing or speech difficulty to practise regularly, build confidence and prepare better for real interviews.

6.1 Limitations

Like every application, InterviewSign has some limitations at the present stage:

1. Internet connection. The application depends on the Supabase cloud for login, data and storage, so it needs an internet connection to work.

2. Fixed question bank. The questions are arranged in predefined rounds, technologies and topics. The user cannot add personal questions.

3. Quality of feedback. The AI feedback and the score depend on the clarity of the answer and, for sign video, on the lighting, the camera position and the quality of the recording.

4. Single language. The interface and the questions are available in English only.

5. Mobile only. The application is made for mobile phones. There is no web version or admin panel for managing questions and users.

6. No human guidance. The feedback is given by the application alone. There is no option to share an attempt with a mentor, a teacher or an interpreter.

7. Basic statistics. The statistics are limited to the number of attempts, the average score and the streak. Detailed charts and topic-wise analysis are not available.

8. Testing scope. The application was tested manually on a physical Android phone and an iOS simulator. Testing on a large number of devices and with many users at the same time is not yet done.

These limitations do not affect the main working of the application. They show the areas in which InterviewSign can grow further, and they are taken up in the next section as future scope.

6.2 Future Scope

The application can be extended in many useful ways. The following enhancements are planned for the future:

1. Regional languages. Hindi, Marathi and other Indian languages can be added for the interface and the questions, so that more users can practise in their own language.

2. Better sign language recognition. Support for Indian Sign Language (ISL) can be improved with a dedicated recognition model for more accurate evaluation of signed answers.

3. Questions in sign language. The questions can be shown by a sign language avatar or a short video, in addition to the text, for users who prefer signs.

4. Detailed progress analysis. Charts of the score over time, topic-wise results and suggestions for weak topics can help the user to know exactly what to improve.

5. Practice reminders. Push notifications can remind the user to practise daily and keep the streak going.

6. Offline mode. Questions can be downloaded to the phone so that the user can practise without internet and the attempts are synchronised later.

7. Admin panel and web version. A web dashboard can let administrators add or change rounds, topics and questions, and let users also use the application from a computer.

8. Mock interviews with mentors. Users can share their attempts with mentors or teachers and take live mock interviews with the help of an interpreter.

9. Career support. A resume builder, job tips and company-wise question sets can be added to make the application a complete career preparation tool.

10. More accessibility options. Dark mode, adjustable font size, high-contrast themes and vibration feedback can make the application comfortable for more users.